

Unsupervised Learning Report

CS7641: Machine Learning

Summer 2026

1 Assignment Weight

The assignment is worth 15% of the total points.

Read everything below carefully as this assignment has changed term-over-term.

2 Objective

The Supervised Learning Report asked which supervised models worked well on Covertypes. The Optimization and Uncertainty in Learning Report asked how training choices affected the PyTorch neural network on Covertypes. This Unsupervised Learning Report asks a different question:

What structure exists in the Covertypes feature space before labels are used, how do dimensionality reduction methods preserve or distort that structure, and how do compressed representations affect clustering and downstream neural-network learning?

This assignment is not just about running clustering and dimensionality reduction algorithms. It is about using unsupervised methods to reason about the geometry of the data. You should investigate whether the feature space naturally separates into groups, whether those groups align with the known `Cover_Type` labels, whether lower-dimensional representations preserve class-relevant structure, and whether dimensionality reduction helps or hurts downstream neural-network performance.

You will use the same Covertypes working dataset from the SL Report. The labels are available, but this is not a supervised clustering assignment. Labels may be used only after unsupervised models are fit, as external evidence for interpretation and evaluation.

You will:

- Reuse the same approximately 20,000-instance Covertypes sample from the SL Report.
- Apply K-Means and Expectation Maximization via Gaussian Mixture Models to the original feature space.
- Apply PCA, ICA, and Randomized Projections to create lower-dimensional representations.
- Re-run clustering after each dimensionality reduction method.
- Rerun the SL PyTorch neural network, using the best justified training recipe informed by your SL and OL Reports, on each dimensionality-reduced representation.
- Analyze whether unsupervised structure aligns with `Cover_Type` labels and whether dimensionality reduction preserves useful structure.

3 Connection to Prior Reports

This assignment continues the same Covertypes sequence used in the SL and OL Reports. You should reuse the same working sample, target definition, preprocessing assumptions, and train/validation/test split strategy where applicable.

The purpose of this continuity is experimental control. If you change the sample, feature handling, preprocessing, split strategy, or target mapping, you must disclose and justify the change. You must also explain how the change affects comparison to your earlier SL and OL findings.

The neural-network portion of this assignment should use the best neural-network configuration and training recipe you can justify from your SL and OL Reports. This may include knowledge gained from the SL model comparison, the OL optimizer ablations, and the OL regularization study. The goal is not to run a new architecture search in UL. The goal is to ask whether dimensionality reduction helps or hurts a strong, previously justified neural-network workflow.

Since dimensionality reduction changes the input dimension, you may adjust the input layer to match the transformed feature space. Apart from this necessary input-dimension change, keep the architecture, optimizer, regularization choices, training budget, splits, metrics, and evaluation protocol as consistent as possible across original-feature and dimensionality-reduced comparisons.

4 Evaluation Overview

The full rubric will be used by TAs during grading, but the table below summarizes the main areas you should expect to be evaluated on. This is not the full rubric. It is meant to help you understand the major categories of the assignment and how to prioritize your work.

The report is not graded only on whether a clustering score is high or whether dimensionality reduction improves neural-network performance. Strong reports are controlled, reproducible, leakage-safe, and interpretive. You should explain what the methods reveal about the Covertype feature space, how those findings relate to `Cover_Type`, and whether compressed representations help or hurt clustering and downstream learning.

Area	Points	What this means
Compliance	16	Correct Covertype task, approximately 20,000-instance sample, proper use of <code>Cover_Type</code> as an external label, IEEE format, Enterprise Overleaf, GT Enterprise GitHub, REPRO instructions, outside sources, AI statement, and legible figures/tables.
Study design	10	Clear preprocessing rationale for mixed continuous/binary features, disciplined use of labels, leakage controls for downstream NN experiments, and at least one clear hypothesis that is later revisited with evidence.
Clustering	14	K-Means and GMM/EM are fit on the original feature space with justified parameter choices, internal clustering metrics, label-alignment evidence, and interpretation of what the clusters appear to capture.
Dimensionality reduction	16	PCA, ICA, and RP are evaluated with method-appropriate diagnostics, including explained variance, non-Gaussianity, distance preservation or distortion, seed sensitivity, and justified component-count choices.
DR + clustering	14	K-Means and GMM/EM are rerun after PCA, ICA, and RP. The report compares original-space and transformed-space clustering and explains how dimensionality reduction changed cluster quality, stability, and label alignment.
NN after DR	12	The best justified neural-network workflow from SL and OL is rerun on PCA, ICA, and RP representations and compared to the original-feature baseline using performance metrics, runtime, and class-level behavior.
Discussion	18	The report synthesizes what unsupervised structure was found, whether it aligned with labels, how method assumptions shaped results, which DR methods preserved useful information, and what limitations affect the conclusions.
Total	100	
Extra Credit	+5	Non-linear manifold learning algorithm explored and reported alongside other dimensionality reduction methods.

A strong report will not treat these areas as disconnected checklist items. The strongest reports build a coherent story, what structure exists in the original feature space, how each dimensionality-reduction method changes that structure, whether those changes help clustering or downstream neural-network learning, and what the results imply about using unsupervised learning on Covertype.

5 Required Dataset for Summer 2026

You will reuse the **Forest Cover Type / Covertypes** dataset from your SL Report. If this dataset is not used, or if `Cover_Type` is not treated as a seven-class multiclass label for external evaluation, you will receive a zero for the assignment.

Forest Cover Type / Covertypes

- **Task context:** Multiclass classification context, used here for unsupervised analysis and downstream evaluation.
- **External label:** `Cover_Type`, classes 1 through 7.
- **Working dataset:** Approximately 20,000 instances sampled from the full dataset, matching the SL Report sample unless a change is explicitly justified.
- **Feature structure:** Mixed continuous cartographic variables and binary wilderness-area / soil-type indicators.

6 Procedure

You will complete one unsupervised learning and dimensionality reduction study on the Covertypes dataset.

You will:

1. confirm that the Covertypes sample and preprocessing choices are consistent with the SL Report,
2. state a hypothesis about clustering structure, dimensionality reduction effects, label alignment, or downstream neural-network behavior,
3. apply K-Means and EM/GMM clustering to the original feature space,
4. apply PCA, ICA, and Randomized Projections,
5. re-apply K-Means and EM/GMM clustering after each dimensionality reduction method,
6. rerun the SL PyTorch neural network on each dimensionality-reduced representation using the best justified training recipe informed by SL and OL, and
7. synthesize what the unsupervised results reveal about the Covertypes feature space and supervised labels.

Hypothesis-driven analysis requirement

You must propose at least **one clear hypothesis** before presenting the main results.

The hypothesis may focus on clustering, dimensionality reduction, label alignment, downstream neural-network behavior, or a relationship among these pieces. A strong hypothesis does not merely say that one method will “perform best.” It predicts a behavior or trade-off and ties that prediction to dataset properties and course concepts.

For example, a strong hypothesis might predict that K-Means clusters will only partially align with `Cover_Type` because Euclidean geometry may capture elevation, distance, and wilderness-area structure rather than ecological cover labels directly. Another strong hypothesis might predict that PCA will preserve downstream neural-network performance better than ICA or RP because Covertypes contains correlated continuous cartographic features and PCA may capture dominant variance efficiently.

You may include more than one hypothesis if it helps your narrative, but only one clear, testable hypothesis is required. Your discussion should explicitly revisit your hypothesis and state whether the evidence supports, contradicts, or qualifies it.

Use of labels in unsupervised learning

The `Cover_Type` labels are available, but they must be handled carefully.

You may use labels to:

- evaluate cluster-label alignment after clustering is complete,
- color visualizations after fitting unsupervised models,
- compute external clustering metrics such as adjusted Rand index or normalized mutual information,
- interpret whether clusters correspond to known cover types,
- evaluate downstream supervised neural-network performance.

You may **not** use labels to:

- fit K-Means, EM/GMM, PCA, ICA, RP, or nonlinear manifold methods,
- choose clusters purely by maximizing agreement with test labels,
- fit dimensionality reduction using validation or test labels,
- tune unsupervised transformations using held-out test performance.

Labels are external evidence for interpretation. They are not inputs to unsupervised fitting.

Preprocessing and leakage controls

Covertypes contains continuous features and binary indicator features. Your preprocessing choices matter.

You must justify how you preprocess the feature matrix before clustering and dimensionality reduction. In particular, discuss whether continuous features are scaled, whether binary indicator variables are scaled or left as 0/1, and how mixed continuous/binary features may affect distance-based methods such as K-Means and PCA.

For exploratory unsupervised analysis, you may fit clustering and dimensionality reduction on the working sample as long as labels are used only after fitting for interpretation. For downstream supervised neural-network evaluation, learned preprocessing and dimensionality reduction must be fit on the training split only and then applied to validation and test data.

This means PCA, ICA, RP, scaling, or any other learned transformation used before neural-network evaluation must be fit without using validation or test data.

General rules

- You may program in Python and are allowed to use standard libraries, **as long as they were not written specifically to solve this assignment**.
- Code repositories, notebooks, or AutoML templates created for CS7641 assignments are **not allowed**.
- TAs must be able to recreate your experiments on a standard Linux machine if necessary.
- The analysis you provide in the report is paramount and the focus of the assignment.

7 Parts and Requirements

Part 1: Clustering on Original Features

Goal: Determine what cluster structure exists in the original Covertypes feature space and whether that structure aligns with the known `Cover_Type` labels.

You must apply the following clustering algorithms to the original feature space:

- **K-Means Clustering**
- **Expectation Maximization via Gaussian Mixture Models (GMM)**

For K-Means, you must justify the distance assumptions, scaling decisions, initialization strategy, number of initializations, random seed handling, and choice of k . Since Covertypes has seven labels, $k = 7$ is a natural reference point, but it is not automatically the correct number of clusters. You should evaluate multiple k values and explain whether the data appears to support fewer, seven, or more clusters.

For EM/GMM, you must justify the number of components, covariance type, initialization strategy, convergence criteria, and random seed handling. You should evaluate multiple component counts and explain whether the mixture model suggests cluster structure that is similar to or different from K-Means.

Required evidence should include:

- at least two internal clustering metrics, such as silhouette score, Davies–Bouldin index, Calinski–Harabasz index, inertia, log-likelihood, AIC, or BIC;
- at least one label-alignment metric or summary, such as adjusted Rand index, normalized mutual information, homogeneity, completeness, V-measure, or a cluster-label contingency table;
- a description of what the clusters appear to capture in terms of feature properties;
- an explanation of whether the clusters align with `Cover_Type` labels and why that alignment may be imperfect.

A good clustering analysis does not assume that good clusters must perfectly match the labels. Clusters may capture elevation, wilderness area, soil type, or distance structure rather than cover type directly. That can still be meaningful if you explain it with evidence.

Part 2: Dimensionality Reduction

Goal: Create lower-dimensional representations of Covertypes and evaluate how each method preserves, distorts, or reveals structure.

You must apply the following dimensionality reduction methods:

- **Principal Component Analysis (PCA)**
- **Independent Component Analysis (ICA)**
- **Randomized Projections (RP)**

For each method, you must justify the number of components selected. You should not choose a component count arbitrarily. Your choice should be tied to explained variance, reconstruction or distortion, non-Gaussianity, runtime, downstream clustering behavior, or downstream neural-network performance.

PCA expectations

For PCA, discuss explained variance and the cumulative variance curve. Report how many components are needed to retain meaningful structure and whether the data appears redundant, low-rank, or strongly correlated.

Useful PCA evidence may include:

- explained variance ratio by component,
- cumulative explained variance,
- reconstruction error,
- rank or effective rank discussion,
- interpretation of whether dominant components appear to reflect feature-scale or domain structure.

ICA expectations

For ICA, discuss whether independent components appear to capture non-Gaussian structure. ICA can be sensitive to preprocessing, component count, initialization, and convergence behavior, so you should report enough detail to make the result interpretable.

Useful ICA evidence may include:

- component kurtosis or another non-Gaussianity measure,
- convergence behavior,
- sensitivity to random seed,
- component-count choice,
- interpretation of whether components appear meaningful or unstable.

RP expectations

For Randomized Projections, discuss whether random low-dimensional projections preserve useful distances or downstream behavior. Since RP is random, you must evaluate variability across multiple random seeds.

Useful RP evidence may include:

- pairwise-distance preservation,
- reconstruction error if you implement an inverse or pseudo-inverse reconstruction,
- projection distortion across random seeds,
- downstream clustering variability,
- downstream neural-network performance variability.

RP is not primarily a reconstruction method unless you explicitly implement reconstruction. If you do not evaluate reconstruction, you should evaluate distance preservation or projection distortion instead.

Cross-method comparison

After evaluating PCA, ICA, and RP individually, compare the representations across methods. Your report should explain why the selected component counts are reasonable for each method and whether the transformed spaces preserve different kinds of structure. For example, PCA may preserve dominant variance, ICA may emphasize non-Gaussian components, and RP may preserve distances approximately while introducing seed variability. A compact table comparing component counts, diagnostics, stability, runtime, and downstream behavior is often more useful than treating each method in isolation.

Part 3: Clustering after Dimensionality Reduction

Goal: Determine how PCA, ICA, and RP change the clustering structure of Covertypes.

You must re-run both clustering algorithms after each dimensionality reduction method:

- K-Means after PCA
- K-Means after ICA
- K-Means after RP
- EM/GMM after PCA
- EM/GMM after ICA
- EM/GMM after RP

This creates six DR + clustering combinations. You do not need to give equal space to every combination, but you must report enough evidence to compare them fairly.

After dimensionality reduction, you should reselect or justify the clustering settings rather than automatically reusing the original-feature settings. For K-Means, this means evaluating or defending the value of k in the transformed space. For EM/GMM, this means evaluating or defending the number of components and covariance structure in the transformed space. Reusing the original-space settings is acceptable only if you explain why that comparison is intentional and scientifically meaningful.

Your analysis should answer:

- Did dimensionality reduction improve or degrade internal clustering metrics?
- Did dimensionality reduction improve or degrade alignment with `Cover_Type` labels?
- Did PCA, ICA, or RP make clusters more stable or more interpretable?
- Did lower-dimensional spaces reveal structure hidden in the original feature space?
- Did dimensionality reduction remove class-relevant information?

Tables are strongly encouraged for this section. A compact table comparing original-space clustering and DR-space clustering can often communicate more effectively than many separate plots.

Part 4: Neural Network after Dimensionality Reduction

Goal: Evaluate whether lower-dimensional representations help or hurt downstream supervised learning when using the best neural-network workflow justified by your SL and OL Reports.

You must rerun the SL PyTorch neural network, using the best justified training recipe informed by SL and OL, with:

- PCA-transformed features,
- ICA-transformed features,
- RP-transformed features,
- and the original-feature neural-network baseline for comparison.

This neural network should be based on the best justified knowledge gained from the SL and OL Reports. For example, your choice may reflect the strongest PyTorch architecture from SL, the optimizer findings from OL, and the regularization findings from OL. You should state what recipe you use and why it is the appropriate baseline for this stage of the course sequence.

Since the input dimension changes after dimensionality reduction, you may adjust the neural network input layer to match the transformed feature dimension. Apart from this input-layer adjustment, keep the architecture, optimizer, training protocol, regularization choices, splits, random seeds, and evaluation metrics as consistent as possible across original-feature and DR-feature comparisons.

For this part, all learned preprocessing and dimensionality reduction transformations must be fit on the training data only and then applied to validation and test data.

Required analysis should include:

- test Accuracy, Macro-F1, balanced accuracy, and class-level behavior;
- fit time and prediction time or training runtime;
- comparison to the original-feature neural-network baseline;
- discussion of whether dimensionality reduction improved runtime, stability, generalization, or minority-class performance;
- explanation of whether performance changes are likely due to denoising, information loss, reduced dimensionality, or altered feature geometry.

A dimensionality reduction method does not need to improve neural-network performance to be interesting. It may reduce runtime, improve stability, preserve performance with fewer features, or reveal that class-relevant information is distributed across many dimensions.

Extra Credit: Nonlinear Manifold Learning

Optional, up to 5 points.

You may implement one nonlinear manifold learning method of your choice, such as UMAP, t-SNE, Isomap, Locally Linear Embedding, or Spectral Embedding. You must justify the method and explain what it contributes beyond PCA, ICA, and RP.

Nonlinear manifold learning can be used for visualization, clustering comparison, or downstream neural-network features, but the use case must be clear. Be careful: some nonlinear methods, especially t-SNE, are primarily visualization tools and do not naturally support standard train/test transformations in the same way as PCA. If you use a nonlinear method for downstream neural-network features, you must explain how train/validation/test transformation is handled without leakage.

Extra credit should include:

- justification of the nonlinear method,
- explanation of key hyperparameters,
- comparison to at least one linear DR method,
- visualization or quantitative evidence,
- and a short discussion of whether the nonlinear representation reveals structure not captured by PCA, ICA, or RP.

8 Required Figures and Tables

The exact layout is up to you, but the main report should include compact evidence for each major claim. You are encouraged to use multi-panel figures and compact tables rather than many separate plots.

Useful figures and tables include:

- class distribution and brief feature/preprocessing summary;
- original-space clustering metrics table;
- cluster-label contingency table or external clustering metric table;
- PCA explained variance curve;
- ICA component non-Gaussianity or kurtosis summary;
- RP distance preservation or seed-variability summary;
- clustering-after-DR comparison table;
- neural-network-after-DR metric and runtime table;
- two-dimensional visualization of selected representations, colored by cluster and/or `Cover_Type`.

Figures and tables must be legible at normal grading zoom. Axis labels, legends, captions, table text, and plotted values should be readable without excessive zooming or guesswork. This is a scored requirement. A small number of clear figures with strong interpretation is better than many crowded figures that cannot be evaluated.

9 Discussion and Conclusion

Your Discussion should synthesize across the full assignment rather than summarize each algorithm separately.

A strong Discussion should address:

- whether Covertypes has natural cluster structure in the original feature space;

- whether clusters align with `Cover_Type` labels, and where they fail to align;
- whether K-Means and EM/GMM reveal different kinds of structure;
- whether PCA, ICA, or RP best preserves useful structure;
- whether dimensionality reduction improves clustering, interpretability, runtime, or neural-network performance;
- whether lower-dimensional representations hurt minority-class behavior;
- whether internal clustering metrics and label-alignment metrics agree or disagree;
- whether your hypothesis was supported, contradicted, or qualified.

You must include a limitations discussion. Good limitations are specific to the study design and evidence. Examples include the approximately 20,000-instance sample, class imbalance, sensitivity of clustering to scaling, instability of ICA or RP, difficulty interpreting clusters in mixed continuous/binary feature spaces, or the fact that `Cover_Type` labels may not correspond to Euclidean cluster geometry.

End with a concise conclusion that explains what unsupervised learning revealed about Covertypes and what a practitioner should take away. For example, your conclusion might state whether dimensionality reduction is useful primarily for visualization, clustering, runtime reduction, denoising, or downstream supervised performance.

10 External Sources and Literature

Include outside sources beyond course materials, with at least **two peer-reviewed references**. These should be used to support your methodological choices or interpretation, not merely listed in the references section.

Primary sources are strongly preferred. Appropriate sources might discuss K-Means, Gaussian mixtures and EM, PCA, ICA, random projections, clustering validation, manifold learning, or evaluation of clustering with external labels.

Avoid relying on textbooks for this requirement. Textbooks can be useful for background, but they are usually too general for supporting specific experimental choices in this report. Avoid using arXiv papers, preprint servers, blog posts, documentation pages, tutorials, lecture notes, Wikipedia, Medium posts, or vendor documentation as one of your required peer-reviewed references.

Use **IEEE citation style**. Cite sources in the body of the paper where they support a claim. Do not only place two papers in the bibliography.

11 Report Requirements

Your report is a formal technical paper limited to **8 pages total**, including figures and references. Anything past 8 pages will not be read.

Your main report must use the **IEEE conference template**. The main report should use the official IEEE conference paper format, such as:

```
\documentclass[conference]{IEEEtran}
```

Your report must be written in L^AT_EX on Overleaf using your **Georgia Tech / Enterprise Overleaf account**. When submitting your report, you are required to include a **READ-ONLY** link to the Overleaf project. Do not share the project directly via email.

Main report versus REPRO

The main report must stand alone as a formal technical paper. It should include the central experimental design, compact figures/tables, key results, and interpretation.

The REPRO PDF may contain extended implementation details, full metric tables, additional clustering results, additional dimensionality reduction diagnostics, full seed summaries, extra plots, and full run logs. These materials support reproducibility and grading, but they do not replace interpretation in the main report.

A good main report does not show everything. It shows the most important evidence and explains what it means.

Interpret, do not summarize

Every figure and table must include a takeaway tied to clustering behavior, dimensionality reduction behavior, label alignment, representation quality, neural-network performance, or runtime trade-offs. Avoid repeating the legend. Explain what the result *means* and why it happened.

For example, do not merely state that PCA retained 95% of variance. Explain whether that retained variance preserved cluster structure, label alignment, downstream neural-network performance, or runtime efficiency.

Formal writing requirement

Your *report* must read like a technical paper, not a checklist. Bullets are allowed when they improve clarity, such as listing hyperparameters, summarizing preprocessing steps, or describing an experimental protocol. However, your **analysis and conclusions must be written in paragraphs** with coherent reasoning.

Excessive bullet-only writing that replaces interpretation, especially in Results, Discussion, or Conclusion sections, will cause the report to be treated as a draft and scored accordingly. Multiple instances of four or more consecutive bullets in analysis-heavy sections should be treated as excessive unless they are clearly part of a compact protocol, hyperparameter list, or reproducibility note.

12 Acceptable Libraries

You may use any standard open-source libraries that support reproducible experimentation and were not written specifically to solve this assignment.

Recommended stack

- **Core ML and evaluation:** `scikit-learn` for clustering, Gaussian mixtures, PCA, ICA, random projections, metrics, preprocessing, and train/test utilities.
- **Neural networks:** `PyTorch` for rerunning the neural network after dimensionality reduction.
- **Data and plotting:** `pandas`, `NumPy`, and `matplotlib`; `plotly`, `altair`, or `seaborn` are also acceptable.
- **Optional nonlinear manifold learning:** `umap-learn` or standard manifold learning tools from `scikit-learn`, if justified.

Library use does not remove the need to explain preprocessing choices, hyperparameters, seeds, metrics, and interpretation.

13 Submission Details

All scored assignments are due by the time and date indicated on Canvas (Eastern Time, ET). Canvas displays times in your local time zone if your profile is set correctly; grading deadlines are enforced in ET.

All assignments are due at 11:59:00 PM ET on the final Sunday of the unit. Submissions received by **7:59:00 AM ET** the next morning are accepted **without penalty**.

Late penalty

After the grace window, the score is reduced **20 points per calendar day** starting at 7:59:00AM ET day-over-day. Treat **11:59 PM ET** as your real deadline, allow time for upload and verification.

What to submit

You will submit **two PDFs**:

1. **UL_Report_{GTusername}.pdf** — your formal IEEE-style report.
2. **REPRO_UL_{GTusername}.pdf** — your reproducibility sheet.

The reproducibility sheet must include:

1. A **READ-ONLY** link to your Overleaf project.
 2. A **GitHub commit hash** from your final code submission.
 3. Exact run instructions to reproduce results on a standard Linux machine.
 4. Environment setup, package versions, commands, data paths, and random seeds.
 5. Confirmation that the Covertypes sample, preprocessing, and splits match the SL Report, or a clear explanation of any changes.
 6. Details for clustering and DR hyperparameters, component counts, random seeds, and metric calculations.
 7. Extended tables, additional plots, and diagnostics that do not fit in the main report.
- Include the READ-ONLY Overleaf link in the report or Canvas submission comment. **Do not send email invitations.**
 - Use the **GT Enterprise GitHub** for course-related code.
 - Personal GitHub repositories are not acceptable for this assignment.
 - Provide sufficient instructions to retrieve code and data. Canvas paths and file names are sufficient when applicable.

Only your **latest submission** will be graded. Please double-check that **both PDFs** are submitted.

14 Feedback Requests

When your assignment is scored, you will receive feedback explaining errors and successes. If you are confused by feedback, start a private thread on Ed.

15 Plagiarism, AI Use, and Proper Citation

The easiest way to fail this class is to plagiarize. **Using the analysis, code, or graphs of others in this class is considered plagiarism.** We care about your analysis: it must be original and grounded in your own experiments.

If you copy any amount of text from other students, websites, or any other source without proper attribution, that is plagiarism. Citing is required but does not permit copying large blocks of text. All citations must use IEEE style for this assignment.

Code from general-purpose libraries is allowed. Code, notebooks, templates, or automated exploration scripts written specifically for CS7641 assignments are not allowed.

LLMs and AI tools

We treat AI-based assistance the same way we treat collaboration with people. You may discuss ideas and seek help from classmates, colleagues, and AI tools, but all submitted work must be your own. The goal of reports is synthesis of analysis, not merely getting an algorithm to run.

Every submission must include an AI Use Statement. List the tools used and what they assisted with and confirm that you reviewed and understood all assisted content.

Allowed with disclosure: brainstorming, outlining, grammar and clarity edits, code generation, code refactoring, and debugging.

Not allowed: submitting AI-written analysis, conclusions, or figures as your own; fabricating results or citations; paraphrasing AI or prior work to evade checks.

Example Statement at the very end of the report before References: “AI Use Statement. I used ChatGPT and Visual Studio Code Copilot to brainstorm and outline sections of the report, generate and refactor small code snippets, debug an indexing issue, and edit grammar and clarity throughout. I reviewed, verified, and understand all assisted content.”

16 Version Control

- v1.0 – 05/19/2026 – TJL refactored UL Report for Summer 2026 Covertypes assignment.

Assignment description written by Theodore LaGrow.